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Morphology and ultrastructure of pharyngeal sense organs of *Drosophila* larvae

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eLife Assessment

In this **fundamental** manuscript, Richter et al. present a thorough anatomical characterization of the Drosophila melanogaster larval pharyngeal sensory system, which is involved in taste-guided behaviors. This study fills a major gap in the larval sensory map, providing a **compelling** neuroanatomical foundation for future investigations into sensory circuits and behavior. The data presented here are of **exceptional** quality and will be of interest to the Drosophila neurobiology community.

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Abstract

This study provides a comprehensive ultrastructural analysis of the pharyngeal sensory system in *Drosophila melanogaster* larvae, focusing on the four major pharyngeal sense organs: the ventral pharyngeal sensilla (VPS), dorsal pharyngeal sensilla (DPS), dorsal pharyngeal organ (DPO), and posterior pharyngeal sensilla (PPS). Our analysis revealed 15 sensilla across these organs, comprising four mechanosensory, nine chemosensory, and two dual-function sensilla. We identified 35 Type I neurons (six mechanosensory and 29 chemosensory) and six Type II neurons with putative chemosensory functions. Additional sensory structures, including papilla sensilla and chordotonal organs in the cephalopharyngeal region, were characterized. This detailed mapping and classification of pharyngeal sensory structures completes their structural characterization and provides a foundation for future anatomical and functional studies of sensory perception in insects. This work represents a significant step towards a complete analysis of the larval sensory system, providing new opportunities for investigating how an organism processes sensory information to navigate and interact with its environment.

Introduction

The larva of the fruit fly (*Drosophila melanogaster*) has emerged as a key model system for investigating how organisms process sensory information (reviewed in Gerber and Stocker 2007; Melcher et al. 2007; Apostolopoulou et al. 2015; Joseph and Carlson 2015; Rimal and Lee 2018; Widmann et al. 2018; Thum and Gerber 2019). Among all sensory modalities, assessing food quality and innocuousness is most critical for survival, particularly in holometabolous insects like *Drosophila*, as its larval stage represents the primary feeding phase during which the organism ingests substantial amounts of food to sustain rapid growth and prepare for metamorphosis (Amrein and Thorne 2005).

By taking advantage of light microscopy, volume electron microscopy and 3D imaging analysis, recent studies provided detailed ultrastructural characterization of the various components of the larval *Drosophila* sensory system (Grueber et al. 2002; Python and Stocker 2002; Caldwell et al. 2003; Fishilevich et al. 2005; Kreher et al. 2005; Hwang et al. 2007; Xiang et al. 2010; Kwon et al. 2011; Apostolopoulou et al. 2014; Stewart et al. 2015; Berck et al. 2016; Larderet et al. 2017; Rist and Thum 2017; Miroschnikow et al. 2018; Hartenstein et al. 2019; Hernandez-Nunez et al. 2021; Richter et al. 2024; Schoofs et al. 2024). These descriptions encompass the peripheral sensory system, including the four cephalic sensory organs — the terminal organ (TO), dorsal organ (DO), ventral organ (VO), and labial organ (LO) — as well as the enteric nervous system, comprising the esophageal ganglion (EG), hypercerebral ganglion (HCG), and proventricular ganglion (PG). Additionally, they include all external body wall sensilla, proprioceptive chordotonal organs, photoreceptors, and most multidendritic sensory neurons that can be found within the larval body segments. In contrast, a detailed ultrastructural description at the level of individual sensilla and cellular resolution is lacking for the pharyngeal sensory system, despite several studies focusing mainly on the general anatomical organization and the mapping of receptor gene expression. This lack of knowledge arises from the technical challenges associated with the three-dimensional visualization of deep, internal structures using electron microscopy—limitations that advancements in volume electron microscopy have now overcome (White et al. 1986; Eberle et al. 2015; Xu et al. 2017; Peddie et al. 2022; Wang et al. 2021; Collinson et al. 2023; Takemura et al. 2023; Winding et al. 2023; Peale et al. 2024).

Accordingly, we have conducted an ultrastructural analysis of the four principal sensory organs of the larval pharynx using volume electron microscopy (EM). These organs include the ventral pharyngeal sensilla (VPS), dorsal pharyngeal sensilla (DPS), dorsal pharyngeal organ (DPO), and posterior pharyngeal sensilla (PPS) (Python and Stocker 2002; Gendre et al. 2004).

Behavioral, physiological, and functional studies have shown that the larval pharyngeal system is instructing taste-guided behaviors. For instance, a single Gr93a positive neuron pair located in the DPS instructs caffeine-dependent choice behavior and aversive odor-caffeine learning (Apostolopoulou et al. 2016; Choi et al. 2016). A current model, therefore, proposes that pharyngeal sensory neurons are activated upon food ingestion (Apostolopoulou et al. 2015; Miroschnikow et al. 2020; Maier et al. 2021). They instruct initiation or repression of food choice, feeding, and associative food learning through direct contact (Choi et al. 2020). In the case of sugars, sensory input from these neurons is combined with internal metabolic state signals, likely conveyed via the enteric system (Mishra et al. 2013; Hückesfeld et al. 2021; Schoofs et al. 2024).

Furthermore, many of the neurons of the DPS, DPO, and PPS are integrated into the pharyngeal organs of the adult, which contrasts with the general principle that adult sensory structures are born during metamorphosis and suggests that the pharyngeal organs may serve a conserved function throughout development (Gendre et al. 2004). Consistent with this, studies in adult

Drosophila have demonstrated that the pharyngeal organs detect both chemical and physical properties of ingested food and contribute sensory feedback essential for coordinating the peristaltic motor program during unidirectional swallowing (Yang et al. 2021; Qin et al. 2024). Pharyngeal receptor neurons responsive to sugars facilitate the continuation of feeding bouts (LeDue et al. 2015; Yang et al. 2021), whereas those activated by bitter compounds or elevated salt concentrations robustly suppress food intake (Kim et al. 2017; Chen et al. 2021; Sang et al. 2024). In addition, pharyngeal mechanoreceptors relay information about food texture, enabling dynamic modulation of feeding rates to balance between complete food rejection and excessive consumption (Joseph et al. 2017; Yang et al. 2021; Qin et al. 2024). Together, these mechanisms coordinate nutrient homeostasis and the avoidance of harmful substances. Therefore, a better understanding of the building blocks that make up the pharyngeal sensory system would not only help to understand how insects organize food-guided behavior but also provide additional opportunities for pest control.

In this study, we define the morphological types of pharyngeal sensilla based on their ultrastructure, utilizing scanning transmission electron microscopy (STEM) of a larval body volume, which enables the automated acquisition of serial sections at high resolution (**Figure 1**). The morphological characterization of the pharyngeal sensory structures is based on six criteria: (a) the presence of a terminal pore; (b) presence of a sensillum shaft; (c) the presence of a cuticle tube; (d) presence of a tubular body; (e) number of dendrites; (f) lamellation of the thecogen cell (**Table 1**). We establish a comprehensive neuron-to-sensillum map for all four pharyngeal organs, identify previously undescribed neurons, and provide a precise classification and unified nomenclature. This work represents a significant step towards a complete analysis of the larval sensory system, providing new opportunities for investigating how *Drosophila* larvae process sensory information to navigate and interact with their environment.

Results and Discussion

General

Pharyngeal sensilla share a common basic configuration typical of insect sensilla. They generally contain single or multiple bipolar Type I neurons (Zacharuk and Shields 1991) surrounded by support cells (Schmidt and Berg 1994; Keil 1997; Klowden 2007; Prelic et al. 2021). In contrast to multidendritic Type II neurons, these neurons extend only one dendrite from the cell body toward the outer cuticle or pharyngeal lumen (Klowden 2007). The dendrites comprise inner and outer dendritic segments, separated by the ciliary constriction (Altner and Loftus 1985). The outer segments are immersed in sensillum lymph, enclosed by the dendritic sheath and thecogen (sheath building) cell (Slifer 1970; Altner and Prillinger 1980; Zacharuk 1980; Steinbrecht 1984; Zacharuk and Shields 1991) (e.g., **Figure 2B**; Figure 2S1-S5 D; Figure 2S5 D; Figure 3S3-S7 C; Figure 4S1-S3 B). Specific structural features indicate sensory function. Mechanosensory Type I neurons typically display a tubular body - a dendritic tip densely packed with microtubules (Thurm 1964; Keil 1997) (e.g., **Figure 2B**; Figure 2S1-S4 E; Figure 3S1-S2). Chemosensory and especially gustatory sensilla are characterized by a terminal pore to the outside or the respective lumen (Slifer 1970; Falk et al. 1976; Altner and Prillinger 1980). This pore is usually formed by the cuticle tube, a structure thicker and less electron-dense than the dendritic sheath, first observed in *Musca domestica* larval sensilla (Chu-Wang and Axtell 1972) (e.g., **Figure 2B**, Figure 2S1-S5 E; Figure 3S1-S7 D; Figure 4S1-S2 C; Figure 4S3 D). The origin of this structure remains unclear, but most likely, it originates from the dendritic sheath and is, therefore, built by the thecogen cell.

Our analysis utilized a STEM volume of a whole first instar *Drosophila melanogaster* larva (Peale et al. 2024) to locate and reconstruct all sensory organs and their sensilla morphology (**Figure 1A**). While the external sensilla and the enteric sensory system were previously described using

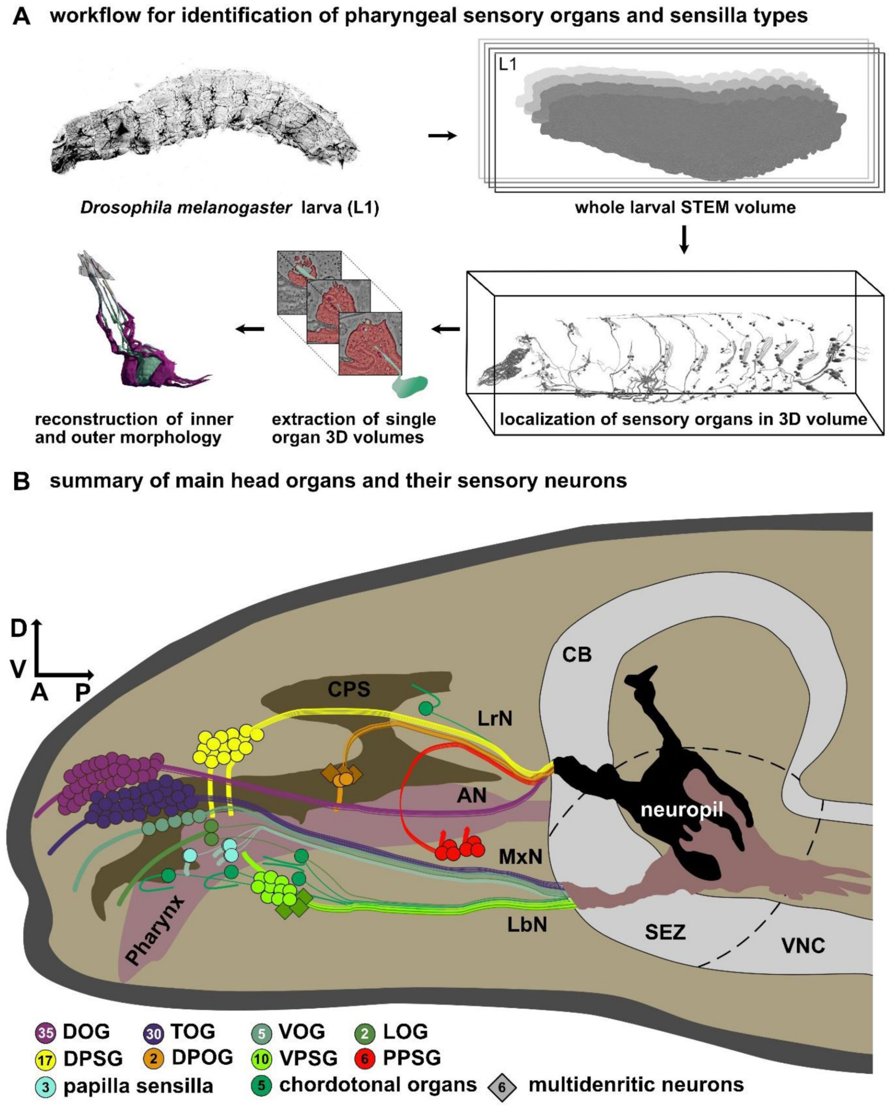

Figure 1.

Localization and analysis of the cellular configuration of the larval pharyngeal sensilla.

(A) Depiction of the workflow for identification of larval pharyngeal sense organs and sensilla types. A previously generated scanning transmission EM (STEM) volume of a whole first instar *Drosophila melanogaster* larva (Peale et al. 2024) was used to localize all cells associated with the larval sensilla using CATMAID (Saalfeld et al. 2009). Smaller volumes were extracted from this volume to generate single organ or sensillum 3D volumes. These volumes were analyzed and manually reconstructed to describe and classify the types of pharyngeal sensilla and their corresponding sensory neurons. **(B)** Schematic drawing of the main external and pharyngeal organs of the larval head region. The projections via the main nerves into the subesophageal zone (SEZ) are visualized. 3D reconstructions of sensilla and sensory neurons were executed from the distal end to the ganglion. Axonal projections were not reconstructed. The numbers in the circles indicate the number of neurons in the respective ganglion. The pharynx (ocher), brain (grey), main external head organs and pharyngeal organs are highlighted. These organs are the dorsal organ (DO; purple), the terminal organ (TO; blue), the ventral organ (VO; light blue), the labial organ (LO; turquoise), the dorsal pharyngeal sensilla (DPS; yellow), the dorsal pharyngeal organ (DPO; orange); ventral pharyngeal sensilla (VPS; bright green) and posterior pharyngeal sensilla (PPS; red).

Abbreviations: D - dorsal; V - ventral; A - anterior, P - posterior, L1 – first instar, CB -central brain, SEZ – subesophageal zone, VNC – ventral nerve cord, AN – antennal nerve, MxN – maxillary nerve, LrN – labral nerve, LbN – labial nerve; CPS – cephalopharyngeal skeleton; DOG - dorsal organ ganglion; TOG - terminal organ ganglion; VOG - ventral organ ganglion; LOG - labial organ ganglion; DPSG - dorsal pharyngeal sensilla ganglion; DPOG - the dorsal pharyngeal organ ganglion; VPSG - ventral pharyngeal sensilla ganglion; PPSG - posterior pharyngeal sensilla ganglion

	VPS					DPS							DPO	PPS	
	P_1	P_2	Pmod	T_1	H_1	T_1	T_2	T_3	T_4	T_5	P/S_1	P/S_2	T_1	T_1	T_2
pore	+	+	-	+	-	+	+	+	+	+	-	-	-	-	-
shaft	+	+	+	-	+	-	-	-	-	-	-	-	-	-	-
tube	+	+	-	+	-	+	+	+	+	+	-	-	+	+	+
tubular body	1	1	1	-	1	-	-	-	-	-	1	1	-	-	-
dendrites	3	3	1	2	1	2	3	4	2	4	1	1	2	3	3
lamellae	+	+	-	+	-	(+)	(+)	(+)	+	+	-	-	(+)	+	+

Table 1

Structural properties of pharyngeal sensilla.

Abbreviations: VPS - ventral pharyngeal sensilla; DPS - dorsal pharyngeal sensilla; DPO - dorsal pharyngeal organ; PPS - posterior pharyngeal sensilla; P – papillum sensillum; Pmod – modified papillum sensillum; T – pit sensillum; H – hair sensillum; P/S – papilla/spot sensillum
+ = structure present; (**+)** = structure weakly present; **-** = structure not present.

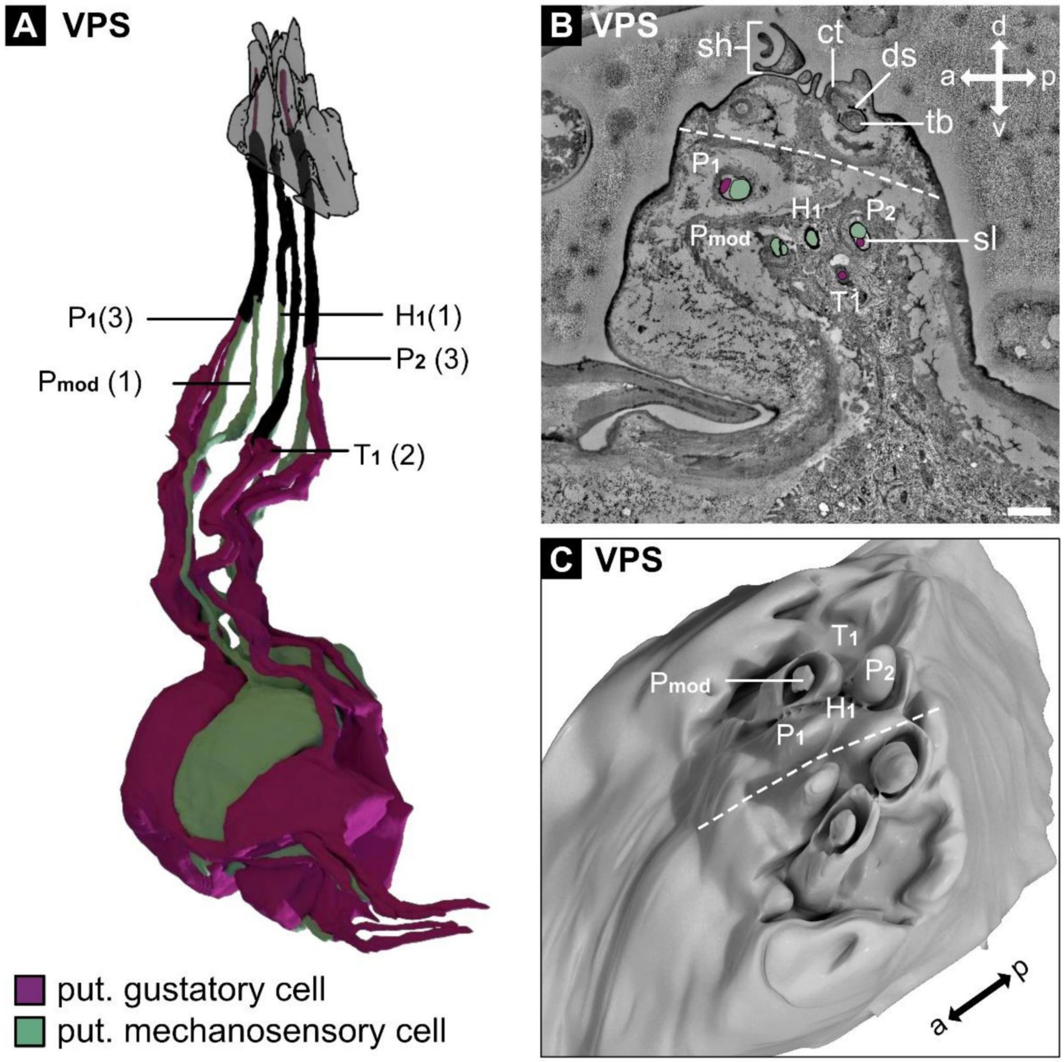

Figure 2.

Main cellular configuration of the ventral pharyngeal sensilla (VPS).

(A) 3D reconstruction of the sensory neurons innervating the VPS. The VPS is a bilaterally organized organ, but the left and right sides are fused. Only the reconstructions of the five sensilla on the left are shown. These are two papillum sensilla (P_1 and P_2), a modified papillum sensillum (P_{mod}), a hair-like sensillum (H_1) and a pit sensillum (T_1). Ultrastructural features of the outer and inner morphology were used to classify the different sensilla. These sensilla resemble canonical sensilla types that can also be found in the terminal organ (see **Figure 1A**), the main external gustatory organ. They display different features like a terminal pore (pit sensillum - T) or a sensillum shaft (modified papillum sensillum - Pmod; hair-like sensillum - H) or both (papillum sensillum - P). Mechanosensory cells are identified by their inner dendritic morphology, which contains a tubular body, a structure known to be important for mechanosensation. For a detailed description of the single VPS sensilla, see supplementary Figures 2 S1 – S5. Nomenclature for sensilla was adapted from previous work (Chu-Wang and Axtell 1972; Rist and Thum 2017). The number of neurons innervating the sensilla is indicated in brackets. (B) STEM section of the VPS showing the organs midline (dashed line). The sensilla of the right side are highlighted, and the innervating neurons are color-coded for their putative sensory function (purple - gustatory; green - mechanosensory). (C) 3D reconstruction of the outer morphology of the VPS. The position of the sensilla is indicated for the right side of the fused organ. Unlike the other pharyngeal sensilla (see **Figure 3** and **Figure 4**), the VPS display prominent outer structures that extend into the pharyngeal cavity.

Abbreviations: d - dorsal; v - ventral; a - anterior, p - posterior, P – papillum sensillum, Pmod - papillum sensillum, H - hair sensillum, T - pit sensillum

Scale bars: (B) 1 µm

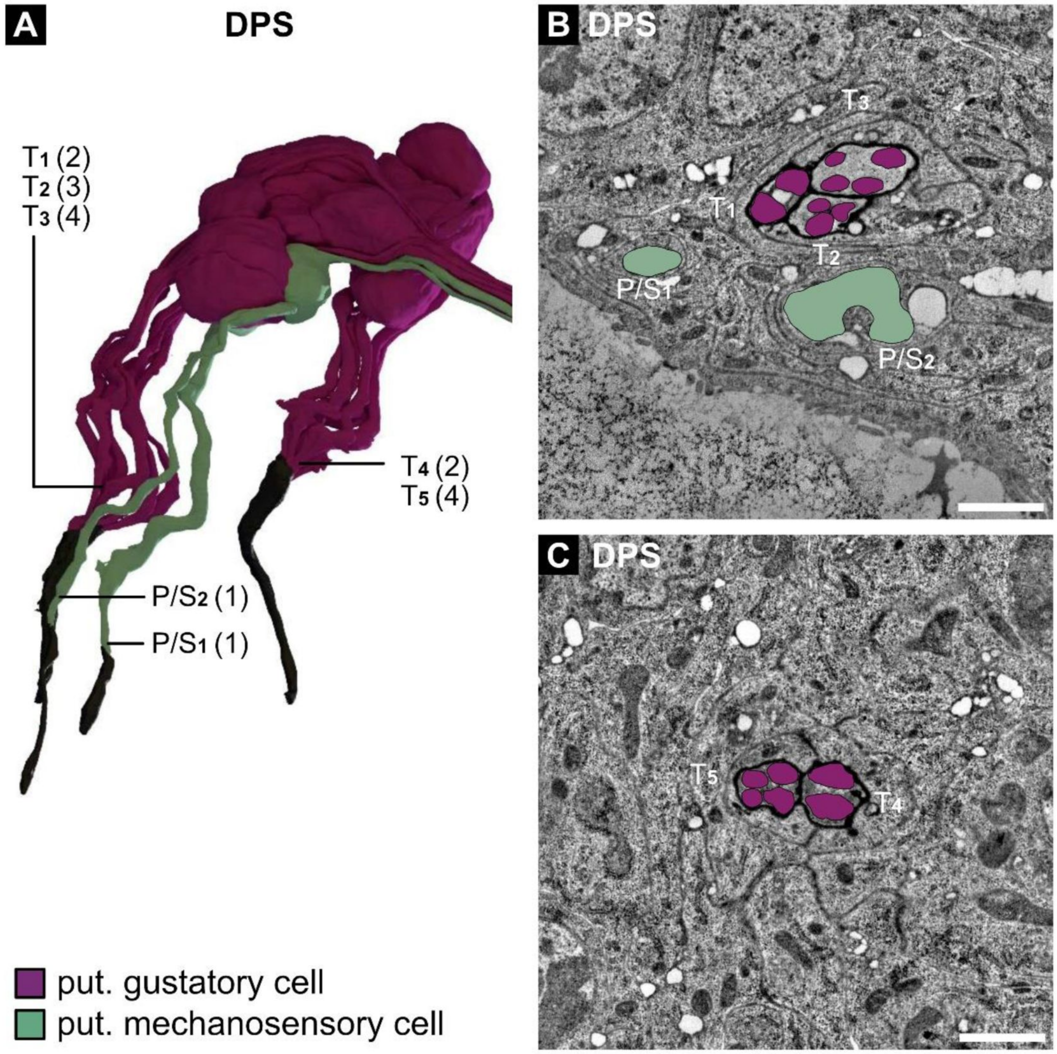

Figure 3.

Main cellular configuration of the dorsal pharyngeal sensilla (DPS).

(A) 3D reconstruction of the sensory neurons innervating the DPS. The DPS is a bilaterally organized organ; only the reconstructions of the seven sensilla on the left are shown. These are two papilla/spot sensilla (P/S_1 and P/S_2) and five pit sensilla (T_1 - T_5). From the ganglion, the sensilla expand to an anterior and a posterior location. In the anterior group, we find three pit sensilla (T_1, T_2 and T_3) and two papilla/spot sensilla (P/S_1 and P/S_2), the latter of which is located in a deep channel. The pit sensilla display an idiosyncrasy as they share their terminal pore into the pharyngeal lumen. The posterior group consists of two pit sensilla (T_4 and T_5), which share a terminal pore, too. Ultrastructural features of the outer and inner morphology were used to classify the sensilla. For a detailed description of the single DPS sensilla, see supplementary Figures 3 S1 – S7. The nomenclature for sensilla was adapted from previous work (Chu-Wang and Axtell 1972; Rist and Thum 2017). The number of neurons innervating the sensilla is indicated in brackets. (B and C) STEM section of the anterior group (B) and posterior group (C) of the DPS. The sensory neurons innervating the sensilla are highlighted and color-coded for their putative sensory function (purple - gustatory; green - mechanosensory).
Abbreviations: d - dorsal; v - ventral; a - anterior, p - posterior, P/S – papilla/spot sensillum, T - pit sensillum
Scale bars: (B) 1 µm; (C) 1 µm

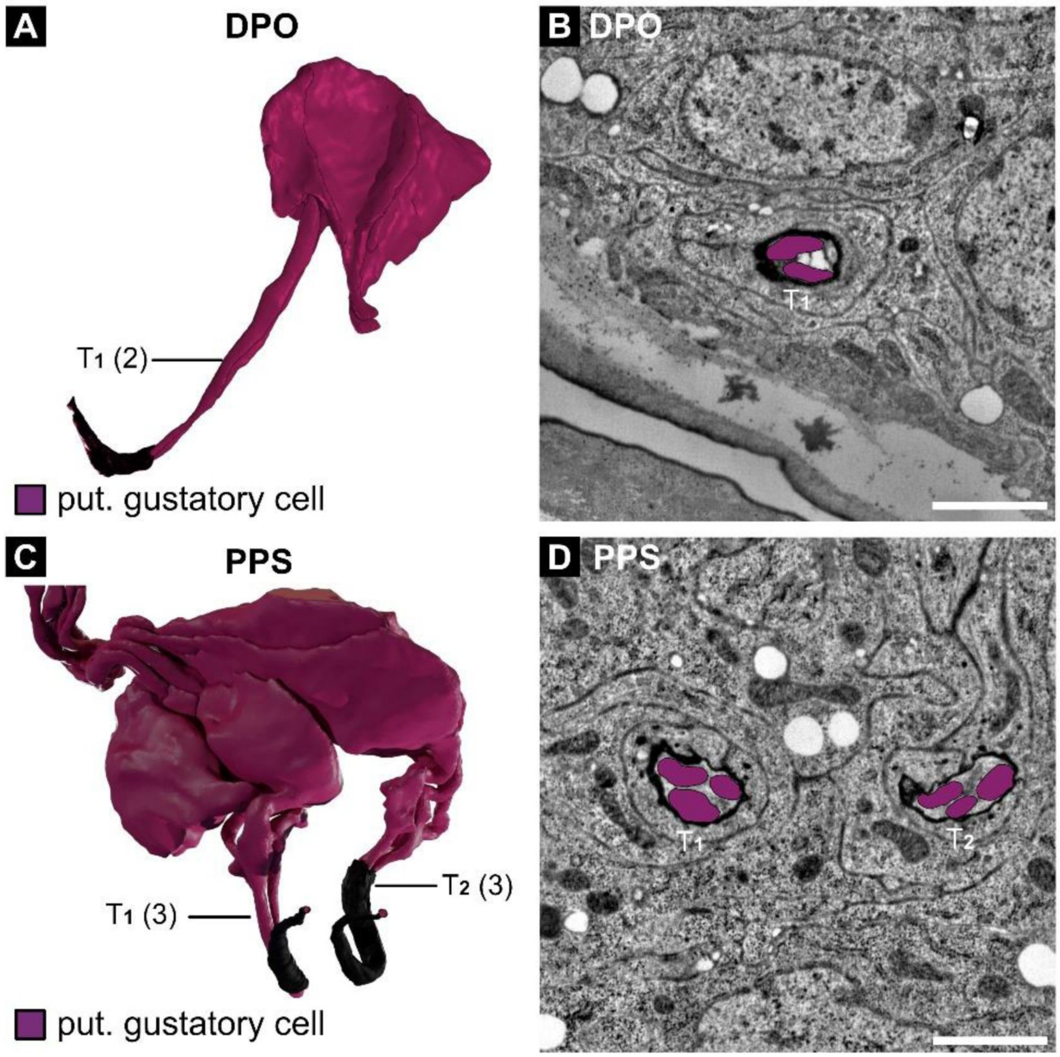

Figure 4.

Main cellular configuration of the dorsal pharyngeal organ (DPO) and posterior pharyngeal sensilla (PPS).

(A) 3D reconstruction of the sensory neurons innervating the DPO. The DPO is a bilaterally organized organ; only the reconstruction of the sensillum on the left is shown. This sensillum is a pit sensillum (T_1) that is innervated by two sensory neurons. These are connected through a pore with the pharyngeal lumen. (B) STEM section of the DPO. The sensory neurons innervating the sensilla are highlighted and color-coded for their putative sensory function (purple - gustatory). (C) 3D reconstruction of the sensory neurons innervating the PPS. The PPS is a bilaterally organized organ; only the reconstructions of sensilla on the left are shown. These sensilla are pit sensilla (T_1 and T_2), innervated by three sensory neurons each. Their pores to the pharyngeal lumen are located in close proximity to each other. (D) STEM section of the PPS. The sensory neurons innervating the sensilla are highlighted and color-coded for their putative sensory function (purple - gustatory). Ultrastructural features of the outer and inner morphology were used to classify the DPO and PPS sensilla. For a detailed description of DPO and PPS sensilla, see supplementary Figures 4 S1 – S3. The nomenclature for sensilla was adapted from previous work (Chu-Wang and Axtell 1972; Rist and Thum 2017).
Abbreviations: d - dorsal; v - ventral; a - anterior, p - posterior, T - pit sensillum
Scale bars: (B) 1 µm; (D) 1 µm

this dataset (Richter et al. 2024; Schoofs et al. 2024), we focus here on the sense organs along the pharynx, which primarily comprise gustatory sensilla that form part of the larval chemosensory system (**Figure 1B**). All pharyngeal organs and identified neurons are arranged bilaterally and thus organized in pairs, which we now describe in detail in the following.

Ventral Pharyngeal Sensilla (VPS)

In contrast to other pharyngeal sensilla, the VPS displays prominent outer structures that extend into the pharyngeal lumen (**Figure 2**). It is a bilateral organ, but the left and right sides are fused (**Figure 2A-C**). Each side contains five individual sensilla innervated by a total of 10 Type I neurons per side (**Figure 2A**). These sensilla resemble sensilla types found in the terminal organ and other external sensilla. It consists of two papillum sensilla (P_1 and P_2) (Figure 2S1-S2), one modified papillum sensillum (P_{mod}) (Figure 2 S3), a hair sensillum (H_1) (Figure 2 S4), and a pit sensillum (T_1) (Figure 2 S5). All VPS sensilla display the standard set of support cells that enclose the sensory neurons and the sensillum lymph (Figure 2 S1-S5 F; Figure 2 S5 E), but the thecogen cells show different degrees of lamellation (**Table 1**; Figure 2 S1-S2 H; Figure 2 S3-S5 G). The papillum sensilla P_1 and P_2 display a shaft with a terminal pore to the pharyngeal lumen (Figure 2 S1-S2 C) and a cuticle tube (ct) (Figure 2 S1-S2 D) that connects with the dendritic sheath (Figure 2 S1-S2 E). Both are innervated by three dendrites each, two that protrude into the pore and, therefore, most likely serve a gustatory function. The third one ends with a tubular body (Figure 2 S1-S2 E) and, therefore, most likely serves a mechanosensory function. P_{mod} displays a shaft that forms a cylindrical portion that encircles a bud-like structure (Figure 2 S3C). It has a single dendrite that ends with a tubular body below the bud. The dendritic sheath remains intact, and a pore is absent. P_{mod} most likely serves a mechanosensory function. H_1 is internally built in a similar configuration as the P_{mod}, containing a tubular body and no pore to the lumen. Only the sensillum shaft is built differently (Figure 2 S4C) and resembles the hair sensilla found in the larval body segments. The different outer morphologies may induce different mechanisms in mechanotransduction or, at least, different stimulus strengths needed (Figure 2 S3-S4 D, E). T_1 resembles a pit sensillum, containing a pore to the lumen and a cuticle tube that connects to the dendritic sheath (Figure 2 S5C-D). The pore channel is innervated by two, most likely gustatory neurons. All cell bodies of neurons innervating the VPS lie in the VPS ganglion (VPSG) (Figure 2 S1-S5I). Furthermore, three multidendritic Type II neurons are present in the VPSG (Figure 2 S3I). All VPSG neurons send axons to the subesophageal zone (SEZ) via the labial nerve (LbN).

Dorsal Pharyngeal Sensilla (DPS)

The DPS is a bilaterally organized organ that contains seven sensilla on each side, innervated by a total of 17 Type I neurons per side. From the ganglion, the neurons extend their dendrites toward anterior and posterior locations in the pharyngeal cavity (**Figure 3 A-C**; Figure 3S1-S7 A, B). In the anterior location, we found three pit sensilla (T_1, T_2, and T_3) and two papilla/spot sensilla (P/S_1 and P/S_2), the latter located in a deep depression (Figure 3S2 B). Papilla and spot sensilla display the same morphology, but the former are termed spot sensilla when found within a compound sensillum (Rist and Thum 2017) and papilla sensilla when occurring as a single sensillum (Dambly-Chaudière and Ghysen 1986; Richter et al. 2024). In the posterior location, we identified two pit sensilla (T_4 and T_5). In contrast to the pit sensilla of other larval gustatory organs, the DPS pit sensilla of the anterior group (T_1, T_2, and T_3) and the posterior group (T_4 and T_5) share a cuticle tube and a terminal pore that opens into the pharyngeal lumen (Figure 3S3-S7 B-C).

Nevertheless, each DPS sensillum possesses its own set of support cells and its own dendritic sheath that separates the dendrites of the sensilla from each other and creates an individual sensillum lymph cavity (Figure 3S3-S7 D-F). The thecogen cells display varying degrees of lamellation, ranging from absent in P/S_1 and P/S_2, to weak in T_1, T_2, and T_3, and pronounced in T_4 and T_5 (Figure 3S1-S7 E-F). All pit sensilla neurons are most likely gustatory, as their dendrite endings are in contact with the pharyngeal lumen and do not display another morphological feature indicating a different sensory modality. The DPS papilla/spot sensilla most likely serve a mechanosensory function, as their dendrite tips contain a tubular body, and the dendritic sheath does not merge into a cuticle tube (Figure 3S1-S2 C-D). Most somata of neurons innervating the DPS lie in the DPS ganglion (DPSG) (Figure 3S2-S7 G). Only the soma of P/S_1 lies nearby outside the ganglion (Figure 3S1 G). All DPS neurons send their axons to the subesophageal zone (SEZ) via the labral nerve (LrN).

Dorsal Pharyngeal Organ (DPO)

The DPO is a small, bilateral, organized structure that contains only one pit sensillum (T_1) on each side, innervated by two Type I neurons (**Figure 4 A-B**; Figure 4S1 A). From the ganglion, the dendrites extend toward a location posterior to the DPS. The pit sensillum features a cuticle tube (Figure 4S1 B) and a terminal pore (Figure 4S1 C) leading to the pharyngeal lumen. Therefore, the two innervating neurons are most likely gustatory. The sensillum includes the standard repertoire of support cells that enclose the sensory neurons and the sensillum lymph (Figure 4S1 D-E). The thecogen cell exhibits weak lamellations (Figure 4S1 F). All cell bodies of neurons innervating the DPO are located in the DPO ganglion (DPOG).

Additionally, three Type II neurons are found in the DPOG (Figure 4 S1G). They extend their multiple dendrites into the hemolymph, where, based on their position, they can perceive internal chemosensory signals. All DPOG neurons send their axons to the subesophageal zone (SEZ) via the labral nerve (LrN).

Posterior Pharyngeal Sensilla (PPS)

The PPS is a bilateral organized structure containing two pit sensilla (T_1 and T_2) on each side that are innervated by three Type I neurons each (**Figure 4C-D**). From the ganglion, the dendrites extend towards the posteriormost location in the pharynx, right before the esophagus (**Figure 1B**). The pit sensilla display a cuticle tube and terminal pore to the pharyngeal lumen. Therefore, the three innervating neurons are most likely gustatory. All cell bodies of neurons innervating the PPS lie in the PPS ganglion (PPSG) (Figure 4S2-S3). All PPSG neurons send their axons to the subesophageal zone (SEZ) via the labral nerve (LrN).

Additional sensory structures

In addition to the main pharyngeal sense organs, we find three single papilla/spot sensilla associated with the feeding apparatus. These resemble classic mechanosensory sensilla found in the larval segments. They possess a tubular body at their dendritic tip, and a dendritic sheath encloses the dendrite. Their dendrites protrude towards the pharyngeal lumen anterior to the DPS/VPS region, and their axons run with the labial nerve (LbN) (Figure 4S4 A-D).

Furthermore, four chordotonal organs are also associated with the larval feeding apparatus (Richter et al. 2024). Three lie along the ventral part of the cephalopharyngeal skeleton (CPS), with the anterior two being monodynal (innervated by a single sensory neuron) and the posterior one being heterodynal (innervated by two sensory neurons). Their axons run with the labial nerve (LbN). The fourth one can be found in the dorsal posterior portion of the CPS. It is also monodynal, and its axon runs along the labral nerve (LN) (Figure 4S4 A, E-H).

In addition to the Type I sensory neurons organized in the sensilla, we also identified six multidendritic neurons associated with the larval feeding apparatus. As mentioned above, three are located within the VPS ganglion, and three are found in the DPO ganglion.

Summary/Conclusion

The larval pharyngeal sensory system comprises four major organs containing 15 sensilla (four mechanosensory, nine chemosensory, two dual-function), with 35 Type I neurons (six mechanosensory, 29 chemosensory) and six Type II neurons in total. Additional sensory structures include three papilla/spot sensilla and four chordotonal organs, distributed across the cephalopharyngeal region, completing our understanding of the larval sensory system. For the first time, the exact sensilla and cell numbers for the pharyngeal organs and sensilla have been described, and a functional prediction of their sensory modality has been made.

In total, 124 Type I neurons are running with the major head nerves (MxN – 47; LbN – 14; AN – 37; LrN – 26). Based on our and previous analyses, there are 55 gustatory or chemosensory neurons, 40 mechanosensory or proprioceptive neurons, 21 olfactory neurons, six thermosensory neurons and two neurons with unidentified function (for details, see **Table 2**). We conclude that these neurons, in conjunction with the enteric system, constitute the primary sensory apparatus for making feeding decisions. This includes navigating towards desirable conditions, evaluating the chemical and physical quality of food, and incorporating this information with the internal state of the animal.

In conclusion, this comprehensive characterization of the *Drosophila* larval pharyngeal sensory system provides an unprecedented mapping of neuronal architecture and function. This detailed neuroanatomical foundation will help to gain new insights into sensorimotor integration, decision-making circuits, and the neural basis of adaptive behaviors in response to environmental changes.

	Sensilla	Olfactory	Gustatory	Chemo (other)	Thermo	Mechano	Other	Total	Ganglion	Nerve
DO	7 OS	21	-	-	-	-	-	**21**	DOG	AN
	3 TS	6	-	-	6	-	-	**6**	DOG	AN
	3 P/S	5	-	-	-	3	2	**5**	DOG	AN
TOdo	1 P	-	3	-	-	-	-	**3**	DOG	AN
	1 Pmod	-	-	-	-	1	-	**1**	TOG	MxN
	1 P/S	-	-	-	-	1	-	**1**	TOG	MxN
TOdi	5 T	-	12	-	-	2	-	**14**	TOG	MxN
	2 K	-	-	2	-	-	-	**2**	TOG	MxN
	3 P	-	8	-	-	3	-	**11**	TOG	MxN
	1 P/S	-	-	-	-	1	-	**1**	TOG	MxN
VO	1 T	-	2	-	-	-	-	**2**	VOG	MxN
	3 P/S	-	-	-	-	3	-	**3**	VOG	MxN
LO	2 P/S	-	-	-	-	2	-	**2**	LOG	MxN
DPS	5 T	-	14	-	-	1	-	**15**	DPSG	LrN
	2 P/S	-	-	-	-	2	-	**2**	DPSG	LrN
DPO*	1 T	-	2	-	-	-	-	**2**	DPOG	LrN
PPS	2 T	-	6	-	-	-	-	**6**	PPSG	LrN
VPS*	2 P	-	4	-	-	2	-	**6**	VPSG	LbN
	1 Pmod	-	-	-	-	1	-	**1**	VPSG	LbN
	1 P/S	-	-	-	-	1	-	**1**	VPSG	LbN
	1 T	-	2	-	-	-	-	**2**	VPSG	LbN
P/S - Ext	1 P/S	-	-	-	-	**1**	-	**1**	-	AN
P/S - Ext	5 P/S	-	-	-	-	**5**	-	**5**	-	MxN
P/S - Ph	3 P/S	-	-	-	-	**3**	-	**3**	-	MxN
ChO-DO	1	-	-	-	-	**1**	-	**1**	-	AN
ChO-TO	1	-	-	-	-	**2**	-	**2**	-	MxN
ChO-CPS	1	-	-	-	-	**1**	-	**1**	-	LrN
ChO-Ph	3	-	-	-	-	**4**	-	**4**	-	LbN
TOTAL	**63**	**21**	**53**	**2**	**6**	**40**	**2**	**124**		
DOG	14	21	3	-	6	3	2	**35**		
TOG	13	-	20	2	-	8	-	**30**		
VOG	4	-	2	-	-	3	-	**5**		
LOG	2	-	-	-	-	2	-	**2**		
DPSG	7	-	14	-	-	3	-	**17**		
DPOG	1	-	2	-	-	-	-	**2**		
PPSG	2	-	6	-	-	-	-	**6**		
VPSG	5	-	6	-	-	4	-	**10**		
single P/S	**9**	-	-	-	-	**9**		**9**		
ChOs	**4**	-	-	-	-	**5**	-	**8**		

Table 2.

Neuronal composition and putative modality of larval head organs and their nerves.

Table updated from Python and Stocker (2002). This summary presents a synopsis of previously reported cell numbers and identities (Singh and Singh 1984; Campos-Ortega and Hartenstein 1985; Schmidt-Ott et al. 1994; Rist and Thum 2017; Richter et al. 2024), as well as the data presented in this work.

* In the VPS and DPS ganglia, there are three cell bodies of multidendritic neurons each, which in sum is in accordance with previously reported cell numbers. Multidendritic neurons (MDNs) have generally not been considered here; however, based on previously reported cell numbers in the head nerves (Miroschnikow et al., 2018), we propose that there are approximately 25 sensory MDNs (per side) innervating the head region through the MxN, AN, and PaN. Additionally, there are 36 sensory MDNs (in total) projecting through the larval vagus nerve (VN) and innervating the aorta, esophagus and gut.

Abbreviations: DO – dorsal organ; TO – terminal organ; TOdo – dorsal group of the terminal organ; TOdi – distal group of the terminal organ; VO – ventral organ; LO – labial organ; DPS – dorsal pharyngeal sensilla; DPO – dorsal pharyngeal organ; PPS – posterior pharyngeal sensilla; VPS – ventral pharyngeal sensilla; P/S – papilla/spot sensillum; ChO – chordotonal organ; OS – olfactory sensillum; TS – thermosensory sensillum; P – papillum sensillum; Pmod – modified papillum sensillum; T – pit sensillum; K – knob sensillum; DOG - dorsal organ ganglion; TOG - terminal organ ganglion; VOG - ventral organ ganglion; LOG - labial organ ganglion; DPSG – dorsal pharyngeal sensilla ganglion; DPOG – dorsal pharyngeal organ ganglion; PPSG – posterior pharyngeal sensilla ganglion; VPSG – ventral pharyngeal sensilla ganglion; CPS – cephalopharyngeal skeleton; Ph – Pharynx; Ext – external; AN – antennal nerve; MxN – maxillary nerve; LrN – labral nerve; LbN – labial nerve

Materials and Methods

Whole larval volume

Sensilla and neuron reconstruction were performed on a STEM (scanning transmission electron microscopy) volume of a whole first-instar larva; technical details of its generation are provided in Schoofs et al. (2024) and Peale et al, 2024. We identified sensory structures by scanning the dataset in the pharyngeal region, thereby recognizing dendritic processes towards the pharyngeal lumen.

Image processing

We extracted smaller volumes of the sensory organs from the entire larval volume. The obtained image stacks were imported into Amira (Thermo Fischer Scientific, v2019). Since the whole larval volume was already aligned, the stacks were only slightly realigned by manual correction. In Amira, structures of interest were segmented and transformed into 3D objects. Next, the segmentations were imported to Blender (Blender Institute, Amsterdam), where the 3D reconstructions were manually finished using the preliminary segmentation as a template.

Data availability statement

The data supporting this study's findings are available from the corresponding author upon reasonable request. The full larval EM volume was published in Peale et al, 2024.

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Additional files

Supplemental Figures

Additional information

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Reviewer #1 (Public review):

Summary:

The authors provide a detailed ultrastructural analysis of the larval pharyngeal sensory organs, including the dorsal pharyngeal sensilla, dorsal pharyngeal organ, ventral pharyngeal sensilla, and posterior pharyngeal sensilla. Using electron microscopy and 3D reconstruction, Richter et al., present a comprehensive mapping and classification of pharyngeal sensory structures, defining mthe orphological type of pharyngeal sensilla based on ultrastructure and generating a neuron-to-sensillum map. These findings significantly advance our understanding of internal larval sensory systems and establish a robust framework for future functional studies in coordination with external sensory systems.

Strengths:

The application of high-resolution electron microscopy and 3D imaging analysis successfully overcomes technical challenges associated with visualizing deep internal structures. This enables an unprecedented level of anatomical detail of the larval pharyngeal sensory system. Thus, the study complements and completes existing maps of larval sensory circuits, contributing a comprehensive neuroanatomical characterization of larval sensory input pathways. These insights will inform future studies on larval behavior, sensory processing, and may also have applied relevance for insect control strategies.

Weaknesses:

While the manuscript is concise, clearly written, and methodologically rigorous, it primarily addresses a specialized readership with expertise in insect neuroanatomy.

https://doi.org/10.7554/eLife.108036.1.sa1

Reviewer #2 (Public review):

Summary:

This manuscript documents the structure of the pharyngeal nervous system of the Drosophila larva. The authors wanted to achieve a detailed ultrastructural reconstruction of the gustatory sensory organs in the Drosophila pharynx. Using serial EM and the associated bioinformatics tools, they have achieved their goal. The paper is written clearly and illustrated beautifully with 3D models and annotated sections. The data will significantly enrich the field of Drosophila neurobiology.

Strengths:

Given the dataset, the findings presented are solid and will be an important work of reference for the future.

Weaknesses:

Previous work, including EM, on the pharyngeal sensory organ is not sufficiently referenced and used for comparison with the data presented in this study.

https://doi.org/10.7554/eLife.108036.1.sa0

Author Response:

We thank the reviewers and editors for their thoughtful and constructive feedback on our manuscript, *"Morphology and ultrastructure of pharyngeal sense organs of Drosophila larvae."* We are pleased that both reviewers found our ultrastructural analysis and 3D reconstructions of the larval pharyngeal sensory system to be of high quality, and we appreciate the recognition of the study's significance and potential impact on the *Drosophila* neurobiology field.

We want to address the concern raised regarding the limited referencing and comparison with previous work on pharyngeal sensory organs, particularly in adult *Drosophila* and other insect species.

As noted by the reviewers, our manuscript is concise and focused. We want to clarify that we initially prepared and submitted this study with the intention of it being considered as a **Short Report**, which comes with limitations on the number of characters and figures that can be included. During the submission process, we were asked by the editors if we would like to submit our work as a full-length **Research Advance**, which we agreed to.

That said, we are now happy to expand the discussion in the broader context of related studies — including prior EM and anatomical work — which would enrich the manuscript and provide readers with a deeper comparative perspective.

We are grateful for the positive assessment of our manuscript and for the opportunity to clarify this point.

Sincerely,

Vincent Richter and Andreas S. Thum

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